Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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DRAFT - team details, declarations, team review and signature pending

1. TEAM AND PROJECT OVERVIEW

Team Name

Us!

Project / Solution Name

Orbyte DecisionOS

Name (Leader)

Awaiting team leader confirmation

Email & Phone (Leader)

Awaiting leader email and phone

University Affiliation(s)

Awaiting university affiliation(s)

Live Prototype URL

https://orbyte-decisionos--jeremyjunxuanta.replit.app/

Code Repository URL (public / access granted to judges)

github.com/jeramibot/orbyte-decisionos (public)

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The Next.js interface sends same-origin API requests to FastAPI. A documented mock Orbyte-style connector retrieves synthetic business inventory and quotations from SQLite through SQLAlchemy. Deterministic Python rules calculate stock coverage, quote totals and delivery-risk comparisons. An owner approves or rejects a stored decision snapshot. Approval, a synthetic purchase-order draft and audit records commit atomically; idempotency prevents duplicate drafts for the same decision.

MiroFish is optional and isolated behind an original HTTP adapter. Persistent simulation requests are claimed by a separate local worker, which records source identifiers, status, errors and genuine completed reports only. Its upstream environment is installed locally at a pinned commit. Small Gemini API verification calls succeeded, but the locally saved Zep value is masked and invalid. No genuine simulation has completed; DecisionOS external execution remains disabled.

The running prototype uses local SQLite and explicitly simulated roles. The public Replit Starter deployment serves the synthetic demo. Its approval and draft workflow persisted after reload on 24 October 2026. It has no real user authentication. Appendix A shows the component boundary.

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2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12 / SQLite	Backend runtime / local storage	PSF / public domain
Next.js 16.4.2; React + React DOM 19.3.1	Connected UI	MIT
TypeScript 5.9.3; Tailwind 4.3.3	Types / styling	Apache-2.0 / MIT
FastAPI 0.135.1; Pydantic 2.12.5; SQLAlchemy 2.0.48	API / validation / persistence	MIT
Uvicorn 0.41.0; HTTPX 0.28.1; dotenv 1.2.2	Server / HTTP adapter / environment	BSD-3-Clause
Other tools and MiroFish	See Appendix B and lockfiles	As listed in Appendix B

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3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
None in active app	Deterministic procurement runs locally	Not applicable
Google Gemini API	Tiny native and compatible endpoint checks; no MiroFish run	AI Studio key; Free tier not confirmed
Zep Cloud	Configured masked value failed authentication	No successful API use
Google Drive / GitHub connectors	Development research and source/template retrieval via Codex	No app runtime dependency
None	Not applicable	Not applicable
None	Not applicable	Not applicable
None	Not applicable	Not applicable

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3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Local macOS	Development and SQLite tests	Local; no cloud region
Replit	Public synthetic demo; approval and draft checked	Starter; Asia; Core voucher expired
GitHub (public)	History and code pushed; CI passed	Owner jeramibot
RunPod	No GPU workload executed	Not applicable
Gemini / Zep Cloud	Tiny Gemini checks; Zep auth failed; no live simulation	Free tier / quotas not verified
None	Not applicable	Not applicable
None	Not applicable	Not applicable

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4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None in active app	No trained/deployed operational model	Not applicable	Not applicable
Gemini 3.5 Flash-Lite	Tiny compatibility test; not in active app	Google AI Studio	Review provider terms
MiroFish scenario model	Optional; no end-to-end run	Zep key pending	No generated result
None	Not applicable	Not applicable	Not applicable
None	Not applicable	Not applicable	Not applicable
None	Not applicable	Not applicable	Not applicable
None	Not applicable	Not applicable	Not applicable

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4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Cafe seed	Project seed.py	Synthetic	Apache-2.0	N
Test fixtures	Project tests	Synthetic	Apache-2.0	N
Scenario inputs	Project scenarios.py	Synthetic	Apache-2.0	N
None	Not applicable	Not applicable	Not applicable	Not applicable
None	Not applicable	Not applicable	Not applicable	Not applicable
None	Not applicable	Not applicable	Not applicable	Not applicable
None	Not applicable	Not applicable	Not applicable	Not applicable
None	Not applicable	Not applicable	Not applicable	Not applicable

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4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex	Code, debugging, tests, design and submission text	Major
Model version	Confirm exact model from team session records	Not asserted here
Whisper base.en / faster-whisper	Local workshop-video transcripts for rules review	Research support
Replit Agent	Automatic import inspection; stopped after 16 seconds / 5 actions	Minor

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4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Codex assisted extensively with implementation, tests, debugging and documentation. Executed checks include 33 backend tests, three component tests, TypeScript checks and a production build. Replit Agent briefly inspected the import before being stopped. Native-browser checks covered approval, rejection, persistence and mobile navigation. Four Playwright tests passed in Replit with installed Chromium. Registered-team review, adaptation and understanding have not yet been confirmed. Before signing, members must inspect the code, reproduce the journey, review assumptions and licence obligations, and record their changes. AI-generated output must not be treated as reviewed merely because automated tests passed.

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5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Free Starter plan; public synthetic demo	Imported, tested and published; proof PDF

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5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was used to import, install, build and test the synthetic prototype on 23 October 2026, then publish and verify its public synthetic demo on 24 October. Proof includes workspace screenshots and passing tests. The team reports its Replit Core voucher expired before redemption; no Core credits are claimed. The application can also run locally on Python and Node.js, so hosting is replaceable. RunPod was not used because no GPU workload was executed. Our mock connector does not use Orbyte production APIs.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application is intended only for invented business, supplier and user records. It stores synthetic inventory, quotations, decision reasons, approvals, draft orders and audit events in SQLite. Free-text fields can accept typed text, so testers must not enter personal or confidential information. The current operational app sends no business data to an external AI provider. Separate tiny Gemini API checks used only test prompts. Enabling MiroFish would send the declared synthetic scenario through the separate service to approved LLM/Zep providers.

Secrets are excluded from Git and never exposed in the frontend. Local data remains until the operator deliberately resets it; no automatic retention schedule exists. Demo roles are not authentication; anyone with the public URL can select an owner role and change shared synthetic records. Real-data use requires real identity, access controls, retention and privacy review. Team contact details in the submission are separate from application data.

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6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

MiroFish has not completed a genuine run, so live compatibility, latency and cost are unverified. Social-agent narratives can be biased, incomplete or mistaken and are not calibrated forecasts. Missing report sections stay empty; failed runs produce an explicit error. Deterministic arithmetic is independent of AI. Requests have time limits and cancellation is best effort; upstream graph/report work can continue. DecisionOS external execution remains disabled; Free-tier eligibility and limits need confirmation before live use.

The operational prototype uses selectable demo roles and local SQLite, with no production authentication, migrations or external purchasing. Delivery and consumption are estimates. Four Playwright tests passed in Replit using installed Chromium; local macOS startup remains restricted. SME validation, production-grade authentication and genuine MiroFish execution remain pending. The free public demo expires after 29 days unless upgraded.

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7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

Awaiting team review, complete project links and leader signature. Readiness boxes are intentionally not marked as confirmed.

DRAFT - team details, declarations, team review and signature pending

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

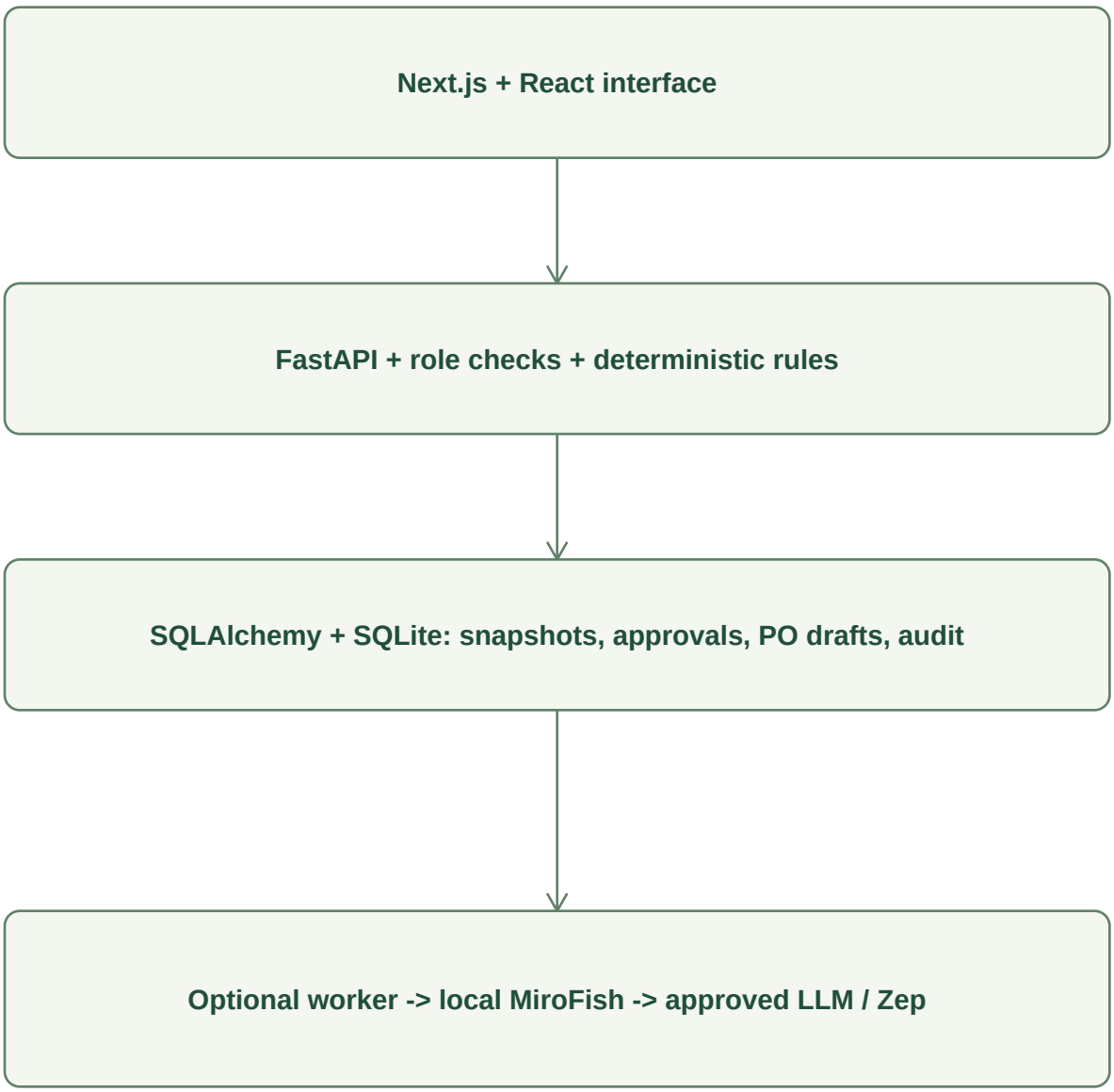
[Signature]

[Date]

Awaiting team review, complete project links and leader signature. Readiness boxes are intentionally not marked as confirmed.

Appendix A: system architecture

Synthetic operational workflow; optional simulation remains disabled.



Our mock Orbyte-style connector supplies business-scoped synthetic inventory and quotes. No official Orbyte API is claimed.

Approval, purchase-order draft and audit records commit together. No real order, payment or inventory receipt occurs.

The public synthetic Replit deployment was verified on 24 October 2026, including owner approval, draft creation and reload persistence. Four Playwright tests passed in the Replit development workspace. See the proof-of-use record.

Appendix B: dependency notices

Python 3.12 / SQLite	PSF / public domain
Next.js 16.4.2 / React 19.3.1 / React DOM 19.3.1	MIT
TypeScript 5.9.3 / Playwright 1.64.1	Apache-2.0
Tailwind CSS / PostCSS plugin 4.3.3	MIT
Lucide React 0.468.0	ISC
FastAPI 0.135.1 / Pydantic 2.12.5 / SQLAlchemy 2.0.48	MIT
Uvicorn 0.41.0 / HTTPX 0.28.1 / python-dotenv 1.2.2	BSD-3-Clause
Pytest 9.1.1 / Vitest 5.0.5 / jsdom 30.1.3	MIT
Testing Library React 16.3.3 / jest-dom 7.0.1	MIT
pytest-asyncio 1.3.0	Apache-2.0
MiroFish 7657031 (separate optional service)	AGPL-3.0

Versions and licences above were checked against installed package metadata and the pinned upstream licence. Full transitive dependency versions are retained in the lockfiles. MiroFish source is not bundled with the Apache-2.0 application. Preserve upstream notices and review AGPL obligations before external deployment.

Development assistance: OpenAI Codex (major use). Whisper base.en via faster-whisper transcribed workshop videos locally for rules review. ReportLab/PyPDF and the Codex presentation runtime generated documents. These research/authoring tools are not application services.